Interacting Multipoles of Rank 1 and 3:

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma,\delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta,\delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta,\gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha,\delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha,\gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha,\beta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha,\beta) \cdot \delta(\gamma,\delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha,\gamma) \cdot \delta(\beta,\delta) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha,\delta) \cdot \delta(\beta,\gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \end{bmatrix} \tag{1}$$

Simplify deltas

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma,\gamma) \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta,\beta) \cdot \mu_\alpha \cdot \Omega_{\beta\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta,\beta) \cdot \mu_\alpha \cdot \Omega_{\beta\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha,\alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha,\alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha,\alpha) \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha,\alpha) \cdot \delta(\gamma,\gamma) \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \delta(\alpha,\alpha) \cdot \delta(\beta,\beta) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha,\alpha) \cdot \delta(\beta,\beta) \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{bmatrix} \tag{2a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \mu_\alpha \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\beta\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\beta\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\delta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{bmatrix} \tag{2b}$$

Simplify R

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{zzz} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\beta\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_z \cdot \Omega_{z\beta\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{bmatrix} \tag{3a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ -15 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{zz\alpha} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_\alpha \cdot \Omega_{\alpha\beta\beta} \end{bmatrix} \tag{3b}$$

Simplify Multipoles

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \end{bmatrix} \tag{4a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\gamma\gamma} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \\ +3 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{y\beta\beta} \end{bmatrix} \tag{4b}$$

Exploit Traceless Conditions

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \tag{5a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \tag{5b}$$

Reduce Indices

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \tag{6a}$$

$$\xRightarrow{\text{cleaned up:}} R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \\ -15 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \tag{6b}$$

Add up

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = R_z^{-9} \cdot -\frac{1}{15} \cdot \begin{bmatrix} -45 \cdot R_z^4 \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \tag{7}$$

Combining everything

$$T_{\alpha\beta\gamma\delta}\mu_\alpha\Omega_{\beta\gamma\delta} = \begin{bmatrix} +3 \cdot R_z^{-5} \cdot \mu_y \cdot \Omega_{yzz} \end{bmatrix} \tag{8}$$